

History-dependent state selection after transient coupling loss in a conductance-based bioelectric network

Supplementary material

Version 1.0 — 15 September 2026

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S1. Scope and equations

This supplement describes existing frozen computations. No new simulation, parameter search or scientific figure was produced for this deposit. The primary endpoints are synthetic electrical states, not anatomy, tissue regeneration or biological memory measurements. The graph has symmetric weights and an unnormalized Laplacian: $L(W) = \text{diag}(W \mathbf{1}) - W$. The coupling term conserves the summed network exchange current; membrane currents and applied stimuli need not conserve the summed voltage.

The conductance-based model is:

$$C_m \dot{V}_i = -G_j[L(W)V]_i - G_L(V_i - E_L) - G_K m_i(V_i - E_K) + g_D a_\infty(V_i) h_i(E_D - V_i) + I_i(t)$$

$$\dot{m}_i = [m_\infty(V_i) - m_i]/\tau_K, \quad m_\infty(V) = \text{logistic}[(V + 0.00076)/0.01409]$$

$$a_\infty(V) = \text{logistic}[(V + 0.025)/0.004].$$

$$\dot{h}_i = [h_\infty(V_i) - h_i]/0.1, \quad h_\infty(V) = \text{logistic}[-(V + 0.040)/0.004].$$

Here $\text{logistic}(x) = 1/(1 + \exp(-x))$. For the persistent variant $h = 1$; for the graded variant $g_D = 0$. The excitable variant uses the inactivation equation above with a 0.1 s time constant. These are independent reduced ODE simulations, not new native BETSE executions.

The cubic model replaces the full membrane-current law and has no potassium gate:

$$\dot{V}_i = \frac{(V_i - V_L)(V_i - V_U)(V_H - V_i)}{\tau_c(V_H - V_L)^2} - \frac{G_j}{C_m}[L(W)V]_i + \frac{I_i(t)}{C_m}$$

S2. Parameters, units and stimulus anchors

Quantity	Value	Meaning
Cm	0.01 F/m ²	Effective membrane capacitance density
GL; GK	1; 2 S/m ²	Leak; potassium conductance density
EL; EK; ED	−0.05; −0.07967507185138671; +0.05 V	Fixed reversal potentials
τK; τh	0.0148; 0.1 s	Constant effective kinetics
Gj	1 S/m ² in central comparison	Common normalization for coupling
gD	4 S/m ² nominal; 3.4, 3.6, 3.8, 4, 4.2, 4.4, 4.6 in validation	Synthetic regenerative conductance
Cubic VL; VU; VH	−0.05; −0.035; +0.01 V	Two stable roots separated by an unstable root
Cubic τc	0.003125 s	Low-state slope approximately −80/s
Persistent anchor	0.082373046875 A/m ²	Earlier original-network upper threshold bracket
Persistent lower bracket	0.08232421875 A/m ²	Earlier development; not independent validation
Excitable anchor	0.12783203125 A/m ²	Separate earlier network anchor
Amplitude factors	0.9, 1.0, 1.1	No validation geometry/cubic recalibration
Pulse	20 ms in regenerative network tests	Applied to one source
Restoration grid	20, 50, 100, 200, 350, 500 ms	Original-geometry validation map
Final validation horizon	5 s; samples every 1 ms	Not proof of unlimited persistence

The potassium midpoint (−0.76 mV) and slope (14.09 mV) derive from Dong et al. (2022), DOI 10.3389/fphar.2022.965086. Their heterologous hKv1.5 measurements do not identify the whole synthetic membrane. The constant τK approximates one reported deactivation timescale. Authenticated Channelpedia data are neither bundled nor used to identify the added regenerative current.

Main temporal tests comprise equal-charge (0.0001 C/m²) and equal-current (0.02 A/m²) series at 1, 10 and 100 ms, τK of 14.8 or 59.2 ms and Gj of 1 or 4 S/m². Exact per-case values are in `protocolo.json`; these graded protocols differ from the later regenerative recruitment assay.

S3. Frozen protocols and enumeration

The main protocol was frozen locally on 14 September 2026 at 21:09:27.908685 UTC, after isolated-cell and original-network development. It contains 487 labels: 144 temporal, 144 memory, 60 conductance-sensitivity, 126 repair, nine no-stimulus and four never-restored conditions. Subsequent controls and corrections remain separately documented. The main freeze was not external preregistration.

Validation was frozen on 15 September 2026 at 13:58:30.658348 UTC. Its 136 unique network cases contribute 65, 49 and 24 block memberships; two cases are shared. Six isolated protocols bring the scientific-condition count to 142. Network numerical verification comprises 136 primary, 136 refined and 44 Radau runs; isolated verification adds 18 runs. These technical runs are not biological replicates.

Original main protocol SHA-256:

5cbb0be57d3d0a06bf0f8ff971c9db2ded15500a959596ed44f1eb7db40680ba.

Original validation protocol SHA-256:

a9d5d8078bfef94c02864a6c7753946882259e0846e44b8541cadb715deac4fb.

The public validation copy changes only its private executable path. ZENODO_PROVENANCE.md and SOURCE_PROVENANCE.json record its distinct public hash and all dependent manifest changes; the original freeze is not attributed to altered bytes.

S4. Configurations and matched histories

Configuration	Source ID	Receivers	Interface edges
historica	61	104	32
fonte_esquerda	50	168	28
fonte_direita	63	170	28
geometria_1701	52	110	37
geometria_2701	61	109	38

All five configurations contain 226 nodes and 622 edges. The two geometry variants use fixed jitter seeds 1701 and 2701 with standard deviation 0.1 times the original median edge length, followed by a Delaunay/minimum-spanning-tree reconstruction retaining 622 edges. Their undirected symmetric edge differences are 12 and 10: six and five replacements, not 12 and 10 replacements. Source variants use the specified coordinate-quantile rule; receiver masks are fixed before outcomes.

Every interface weight is reduced from 1 to 0.1 until restoration. Original-interface loss is $32 \times 0.9 = 28.8$ weight units. Random controls match edge count and per-edge loss within each configuration. Exact restoration reinstates every original weight without resetting voltage or gates. Once the pulse ends, paired histories have identical final W , membrane equations, parameters and zero external input. Different trajectories can therefore reach distinct states under the same final vector field.

S5. Hysteresis and stability checks

The graded model has one zero-current equilibrium at -51.497483 mV with largest real eigenvalue $-86.441007/s$. Its loop areas at 0.2, 1 and 5 s per leg are 0.0032821472, 0.0006794626 and 0.0001363633 V A/m². Slower-loop shrinkage and convergence to a common rest support kinetic lag.

Persistent-model zero-current roots are -50.971228 , -36.016572 and $+7.547151$ mV, with largest real eigenvalues -79.219846 , $+337.710679$ and $-104.270068/s$. The 1 and 5 s loop areas are 0.0247225890 and 0.0238513057 V A/m², with switching in both directions. The 0.2 s ramp does not switch equivalently. Loop area alone is not proof of quasistatic hysteresis or network memory; the current sweep deliberately reaches nonphysiological voltages.

The intact and restored spatial witnesses, case0147 and case0159, have full-network largest real eigenvalues -59.816278 and $-80.906415/s$. They retain 1 versus 226 high nodes and 0 versus 104 high receivers. Recorded residuals are 1.15×10^{-12} and 1.81×10^{-8} V/s; separate small-perturbation checks returned to these states. These checks support local stability and do not map global attraction basins.

S6. Repair and excitable-model results

Repair restores 4, 8 or 16 interface edges before stimulation. The directed ranking uses the integrated absolute voltage difference across edges during an earlier graded response; three fixed random orders are comparators. In all 27 graded comparisons, the directed peak exceeds the random peak by 0.00810–0.04001 mV. At persistent-model factor 0.9, directed repairs leave 0/104 receivers high whereas the sampled random repairs recruit 104/104. At higher tested amplitudes both strategies are collective. This is a heuristic endpoint-dependent ranking, not an optimal repair theorem (Figure S3).

The inactivating extension shows transient excitability in isolated tests and returns to rest. A 2 ms suprathreshold isolated pulse produces a delayed large excursion; two-pulse tests distinguish short-interval suppression from recovery. In the original network, restoring the interface at 20 or 50 ms permits large receiver excursions, whereas restoration at 200 or 500 ms does not in the corresponding weak-interface comparisons. Some other conditions have small threshold crossings without large regenerative excursions. The bottom row of Figure S1 counts threshold crossings and must not be interpreted as a spike count. No persistent attractor memory is inferred for the excitable variant.

S7. Random, never-restored and ablation controls

At the lowest fixed stimulus, random weakening recruits collectively in four of the five validation configurations; the right-source configuration remains localized. The central response is therefore not exclusive to the chosen interface.

Five persistent never-restored interface controls end with 122, 58, 56, 226 and 117 high nodes, respectively. Their high-receiver counts are 0, 0, 0, 110 and 0. One geometry thus recruits without restoration, limiting any claim that restoration is necessary for propagation. Never-restored conditions cannot themselves establish history dependence under identical final connectivity. A completely cut interface cannot transmit across the missing path; recruitment after reconnection is a different event.

Validation includes six no-stimulus cases and twelve additional-current ablations. No-stimulus departure is at most 6.94×10^{-18} V. Ablations return to rest, although some trajectories cross the operational threshold transiently. The cubic ablation leaves pure leak, whereas conductance-model ablation removes the additional depolarizing current and retains leak plus potassium dynamics.

S8. Negative cubic result and limits of transfer

The cubic model has roots -50 , -35 and $+10$ mV with local eigenvalues -80 , $+60$ and -240 /s. Four perturbed zero-current holds and two ramps (1 and 5 s legs) verify bistability and hysteretic switching before the network tests. Despite source activation, all 21 stimulated network conditions return to rest without receiver recruitment. All twelve paired intact/restored comparisons have no history-dependent endpoint contrast; their maximum voltage difference is 6.94×10^{-18} V.

Bistability and hysteresis alone were insufficient to produce network-level history dependence under the fixed stimulation protocol. The comparison did not match the regenerative current shape, collective threshold or basin accessibility. No cubic retuning was performed. This is a protocol-specific negative result, not proof that scalar cubic networks can never exhibit history dependence.

S9. Numerical verification and effective outputs

The historical main verification initially failed six cases at a 0.02 mV criterion, with maximum discrepancy 0.0317561 mV. Their original files remain. Separately labeled replacements in `precisao_corrigida` reduced the maximum refined comparison to 0.000210172 mV; Radau checks retained the classifications. Effective results are `metricas_conferidas.csv` and `resultados_conferidos.jsonl`. The historical failed `verificacao.json` must not be mistaken for the final correction status.

All 136 validation conditions passed refinement without replacement. Maximum primary/refined difference was 0.000214893 mV; maximum Radau/refined difference was 1.115049×10^{-11} V in 44 prespecified comparisons. Effective validation trajectories are in `refinadas`, linked by `effective_source` in `resultados_conferidos_validacao.jsonl`. All accepted validation endpoints satisfy max voltage derivative $<10^{-5}$ V/s, gate derivative $<10^{-4}$ /s where applicable, and full-Jacobian maximum real eigenvalue $<-10^{-6}$ /s. History contrast additionally requires >1 mV paired endpoint difference. Recruitment uses the model-specific intermediate root and a $\geq 50\%$ receiver threshold; it is not an action-potential definition.

Primary DOP853: `rtol` 10^{-10} , `atol` 10^{-12} , max step 2.5 ms. Refined DOP853: 10^{-12} , 10^{-14} , 1 ms. Radau: 10^{-11} , 10^{-13} , 2.5 ms. Event boundaries are explicitly segmented. The completed validation audit rechecked all 136 trajectories and 87 paired comparisons. During deposit preparation, read-only recomputation again agreed with stored receiver counts and peaks; all 959 source scientific files retained their recorded hashes.

S10. Reproduction and provenance

The ZIP preserves both frozen scientific trees and their two required inputs. Full trajectories are retained to permit independent endpoint, transient and solver audits, including the six failed original main cases; compressed arrays are stored once rather than duplicated in display-oriented folders. Exploratory calibration records explain the origin of the fixed anchors and are historical evidence, not new validation. Private literature caches, authenticated datasets, environments, Git history and journal-planning files are excluded.

Original scientific environment: Python 3.9.6, NumPy 1.26.4, SciPy 1.13.1, Matplotlib 3.7.5. Dependencies are specified, not redistributed. Use `verify_bundle.py` after extraction for byte-level checks without model execution. Reproduction scripts prepare new sibling directories by default and run the archived protocol only with the explicit `--run` option. Main reproduction reruns the 487-case experiment and numerical correction, not every historical exploratory or supplementary procedure; final validation reproduction includes the 136 network and six isolated protocols. Figure-only scripts and the full scientific scripts are provided separately. Consult `README_ZENODO.md` for exact commands and scope.

Figure S1

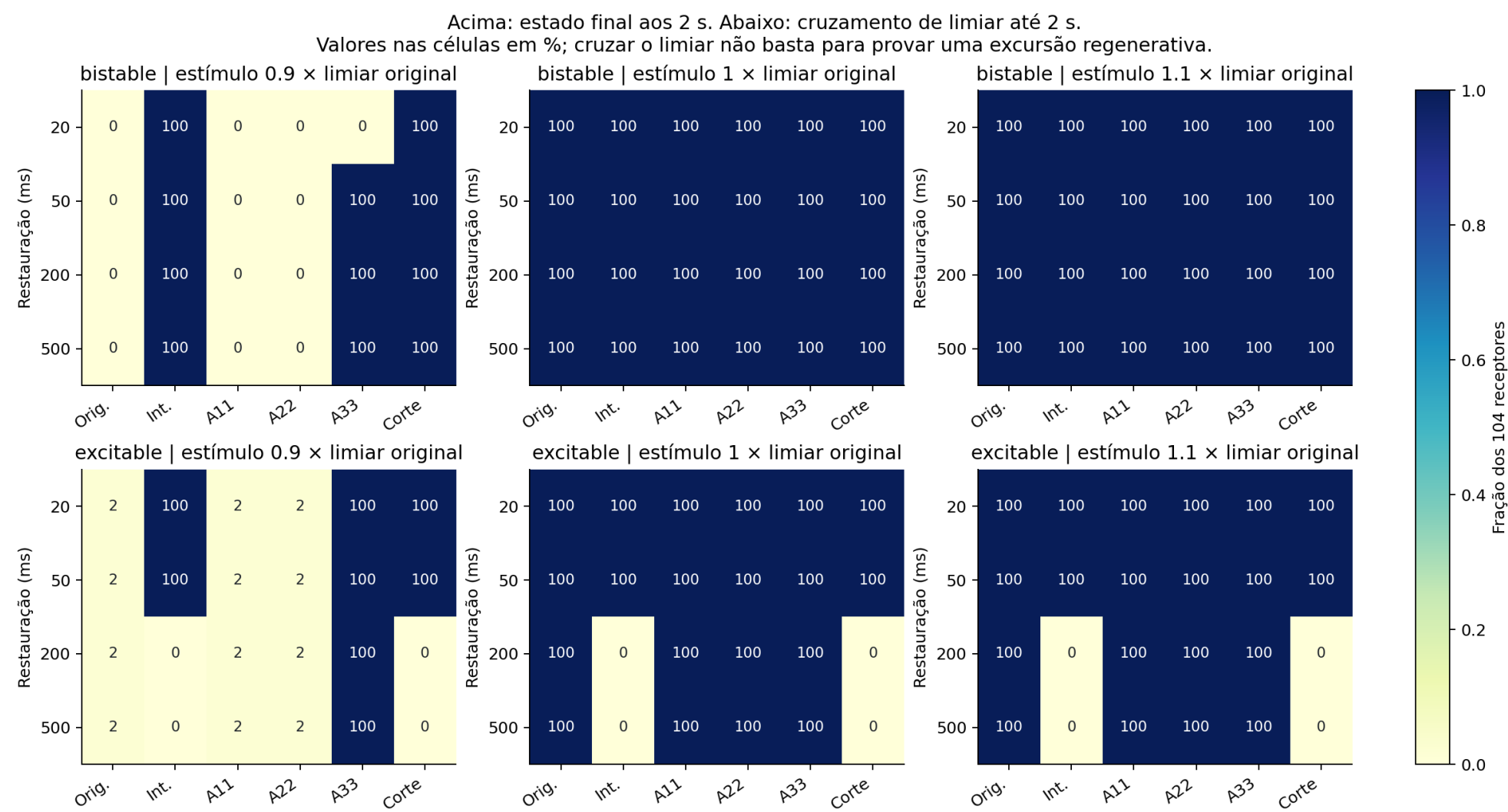


Figure S1. Main-campaign recruitment by restoration time and amplitude. Top: persistent-model final high-receiver fraction at 2 s; bottom: excitable-model fraction ever crossing the fixed threshold by 2 s. Color scale is a fraction, cell annotations are rounded percentages. Crossing that threshold alone is not evidence of a regenerative excursion. Orig. = intact; Int. = 10%-weight interface; A11/A22/A33 = sampled random weakenings; corte = complete cut followed by restoration. Restauração = restoration, in ms. Original source/receiver masks are fixed.

Figure S2

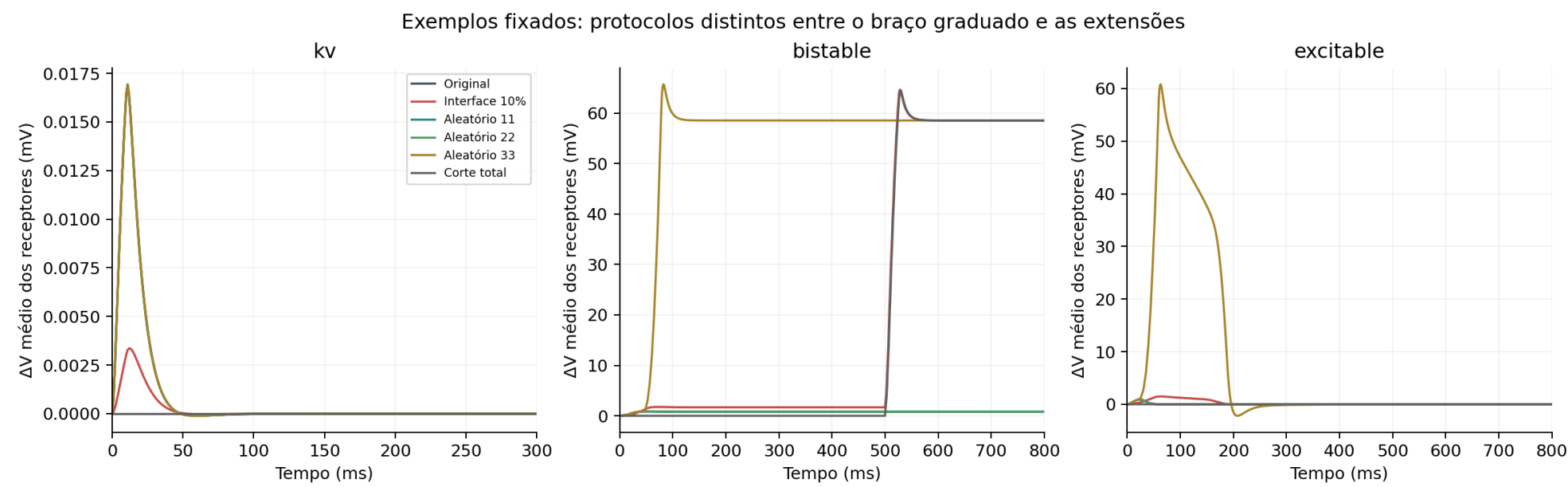


Figure S2. Archived example receiver responses for graded, persistent and excitable variants. Axes: time (ms), mean receiver voltage change relative to rest (mV). Protocols differ between the graded and regenerative examples, as specified in the original figure script; panel heights must not be interpreted as a dose-matched effect size. Interface 10% denotes nonzero coupling; corte total denotes a complete cut followed by scheduled restoration. Original = intact; aleatório = random.

Figure S3

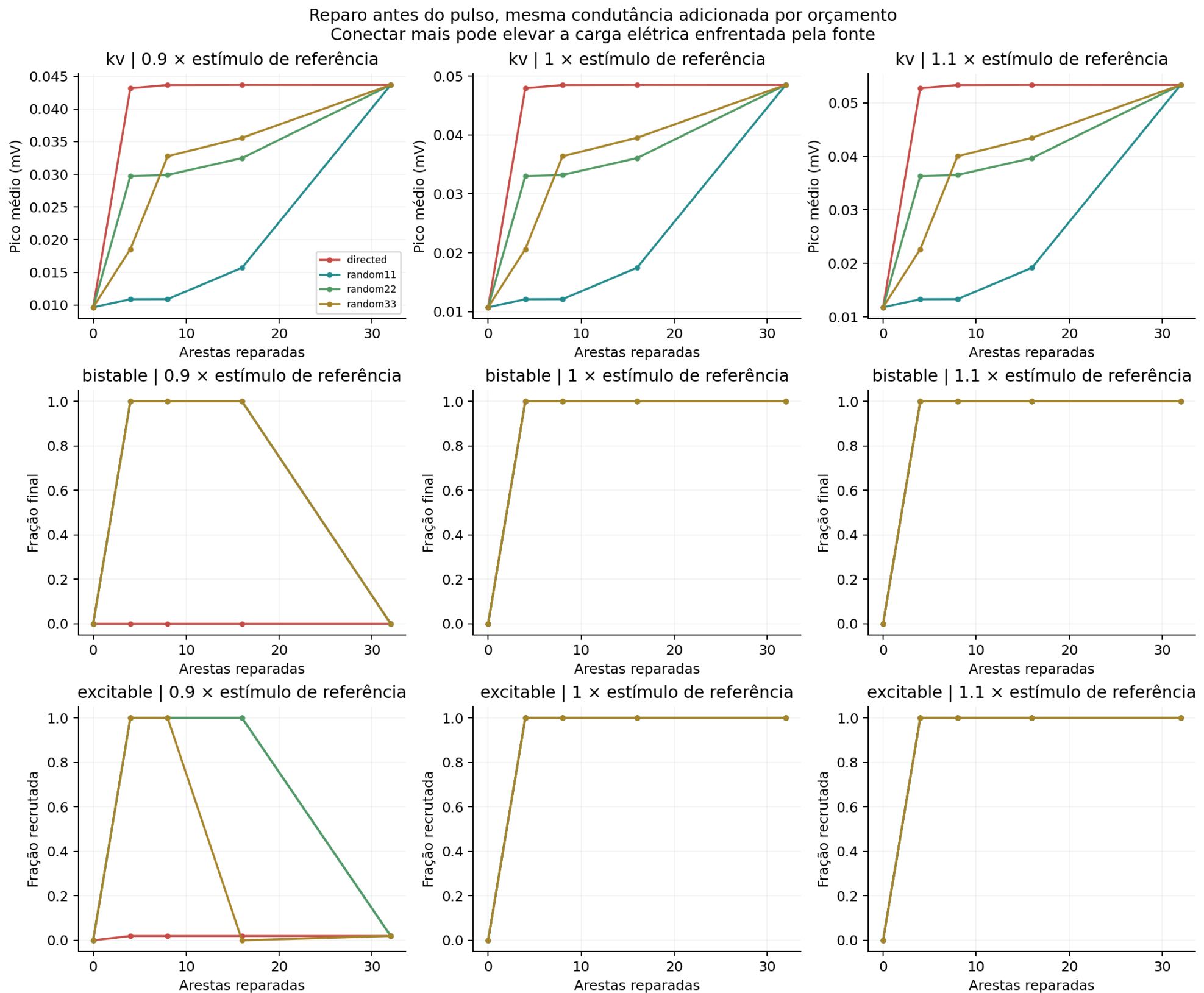


Figure S3. Partial repair before stimulation, a separate assay from complete post-perturbation restoration. Columns: amplitude factors 0.9, 1 and 1.1. Rows: graded peak (mV), persistent final fraction, excitable ever-crossing fraction. Horizontal axis: repaired interface edges; 4, 8 and 16 are tested partial budgets and endpoints show damaged/intact references. Directed repair is compared with three fixed random orders. At factor 0.9, the graded ranking does not preserve persistent recruitment ordering. Fração final = final fraction; fração cruzada = fraction ever crossing. Line segments guide the eye and do not supply untested budgets.